

KnowSelf

Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

Enables agents to autonomously regulate knowledge utilization through situational self-awareness.

CORE MECHANISM

3 modes

fast · slow · knowledgeable

PEAK ALFWORLD

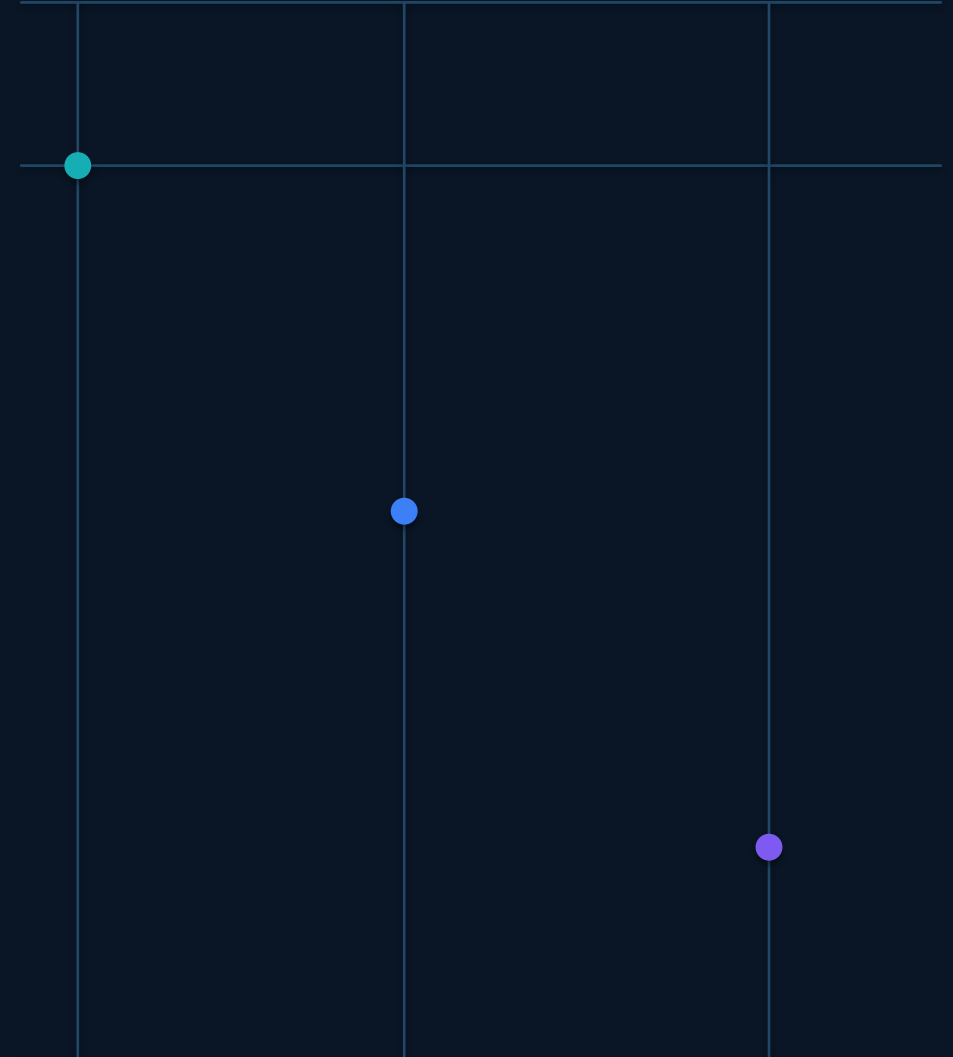
84.33%

Llama-8B KnowSelf

KNOWLEDGE USE

15.01%

selective external lookup



Current Agent Training Uses ‘Flood Irrigation’

Existing methods inject supervision or knowledge without asking whether the current decision step needs it.

ReAct

Trajectory imitation

- Learns from gold trajectories
- Can collapse on unexpected inputs
- No step-level self-assessment

same intervention regardless of
situational need

Reflexion

Trial-and-error refinement

- Adds external feedback after failure
- Spends compute on broad reflection
- Does not identify knowledge need

same intervention regardless of
situational need

KnowAgent · WKM

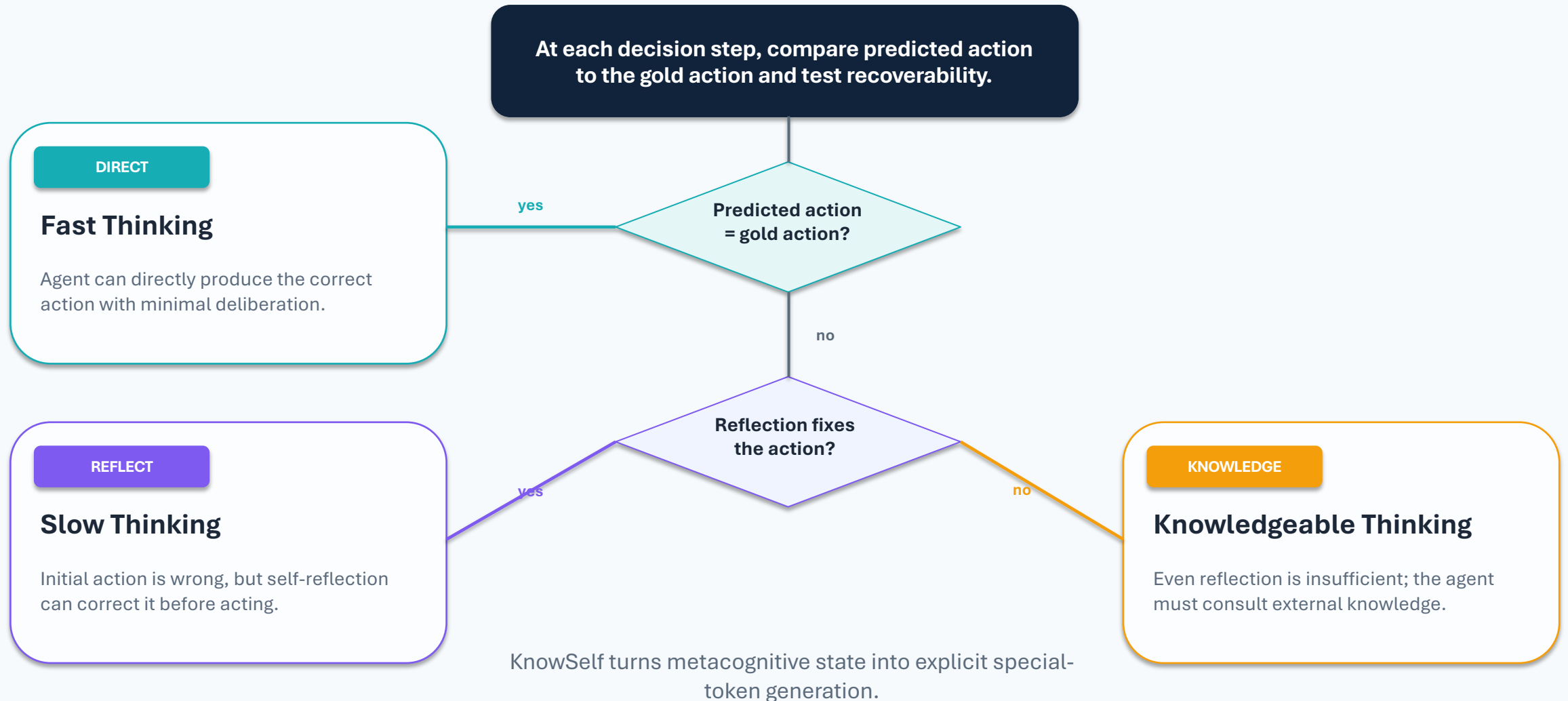
Knowledge-augmented planning

- Consults domain knowledge aggressively
- High inference cost
- Knowledge can hurt when unnecessary

same intervention regardless of
situational need

“Flood irrigation” ⇒ fragile planning, pattern collapse, and unnecessarily high inference costs

Three Situational Modes: Fast, Slow, Knowledgeable



KnowSelf Framework: Data → Training → Inference

1

Self-Awareness Data Construction

Self-explore environments; use heuristic situation judgment; insert special tokens.

2

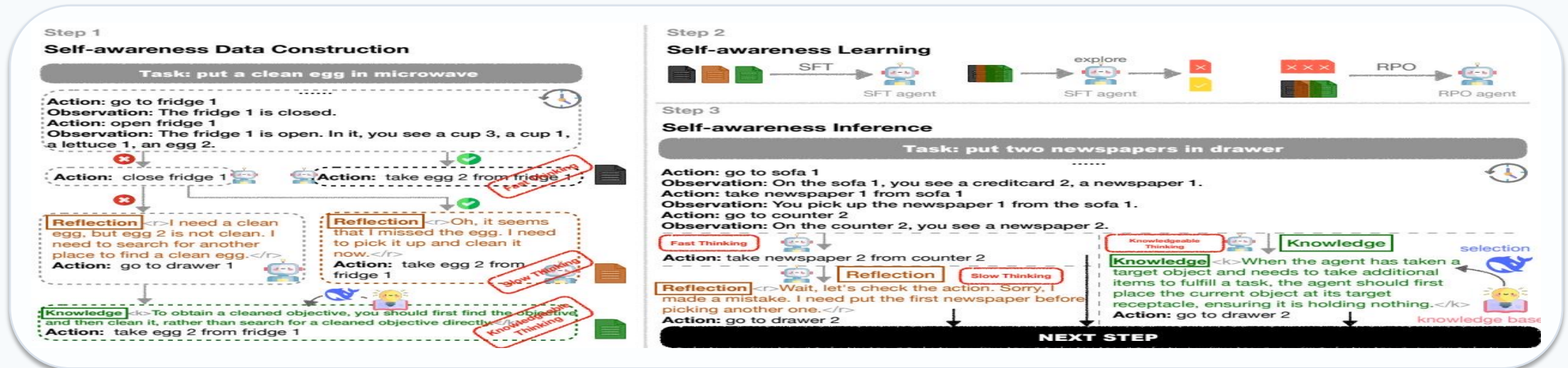
Two-Stage Training

SFT teaches initial self-awareness patterns; RPO refines token selection.

3

Self-Aware Inference

Generated tokens trigger direct action, reflection, or external knowledge retrieval.



Framework figure reference: data construction, learning, and self-aware deployment pipeline.

Qualitative Examples: Reflection and Knowledge at the Right Time

(a) Task: put a hot mug in cabinet	(b) Task: put a saltshaker in drawer
OpenAI-O1 Action: take mug 1 from countertop 1 Observation: You pick up the mug 1 from the countertop 1. Action: go to microwave 1 Observation: The microwave 1 is closed. Action: open microwave 1 Observation: You open the microwave 1. In it, you see an egg 1. [Knowledge] <k>If the agent is in a situation where it needs to modify the temperature of an object using a receptacle (e.g., a fridge for cooling), it should directly perform the specific action like 'cool' or 'heat' instead of placing the object on it.</k> Action: take egg 1 from microwave 1 Observation: Nothing happens.	KnowSelf (Llama-8B) Action: take mug 1 from countertop 1 Observation: You pick up the mug 1 from the countertop 1. Action: go to microwave 1 Observation: The microwave 1 is closed. Action: open microwave 1 Observation: You open the microwave 1. In it, you see an egg 1. Action: put mug 1 in/on microwave 1 [Reflection] <r>Wait, let's check the action. Let's think step by step. With the microwave open, I can heat the mug using the microwave.</r> Action: heat mug 1 with microwave 1
DeepSeek-R1 Action: take saltshaker 1 from countertop 3 Observation: You pick up the saltshaker 1 from the countertop 3. Action: go to drawer 1 Observation: The drawer 1 is open. In it, you see a fork 1. Action: put saltshaker 1 in/on drawer 1 [Reflection] <r>Wait, the drawer is open, so the saltshaker might fall inside. I need to close the drawer first.</r> Action: close drawer 1 Observation: You close the drawer 1. Action: put saltshaker 1 in/on drawer 1 Observation: Nothing happens.	KnowSelf (Llama-8B) Action: take saltshaker 1 from countertop 3 Observation: You pick up the saltshaker 1 from the countertop 3. Action: go to drawer 1 Observation: The drawer 1 is closed. Action: open drawer 1 Observation: You open the drawer 1. In it, you see a fork 1. [Knowledge] <k>When the agent encounters a situation where receptacles contain other items but only needs to place the target object, the agent should proceed to place the object without removing unrelated items.</k> Action: put saltshaker 1 in/on drawer 1

REFLECTION TOKEN

Task: put a hot mug in cabinet

- OpenAI-O1 opens the microwave, then incorrectly tries to take an egg.
- KnowSelf reflects that the mug must be heated using the open microwave.

Situational read: wrong first action, recoverable by self-correction.

KNOWLEDGE TOKEN

Task: put a saltshaker in drawer

- DeepSeek-R1 gets stuck trying to close the already-open drawer first.
- KnowSelf uses knowledge: place the object without removing unrelated items.

Situational read: reflection insufficient, external knowledge needed.

ALFWorld Results: Strong Accuracy with Selective Knowledge

LLAMA-8B KNOWSELF

84.33%

GEMMA-2B KNOWSELF

79.85%

BLANKET-KNOWLEDGE BASELINES

100%

best Llama-8B overall: 15.01% knowledge

matches GPT-4o ExpeL: 26.41% knowledge

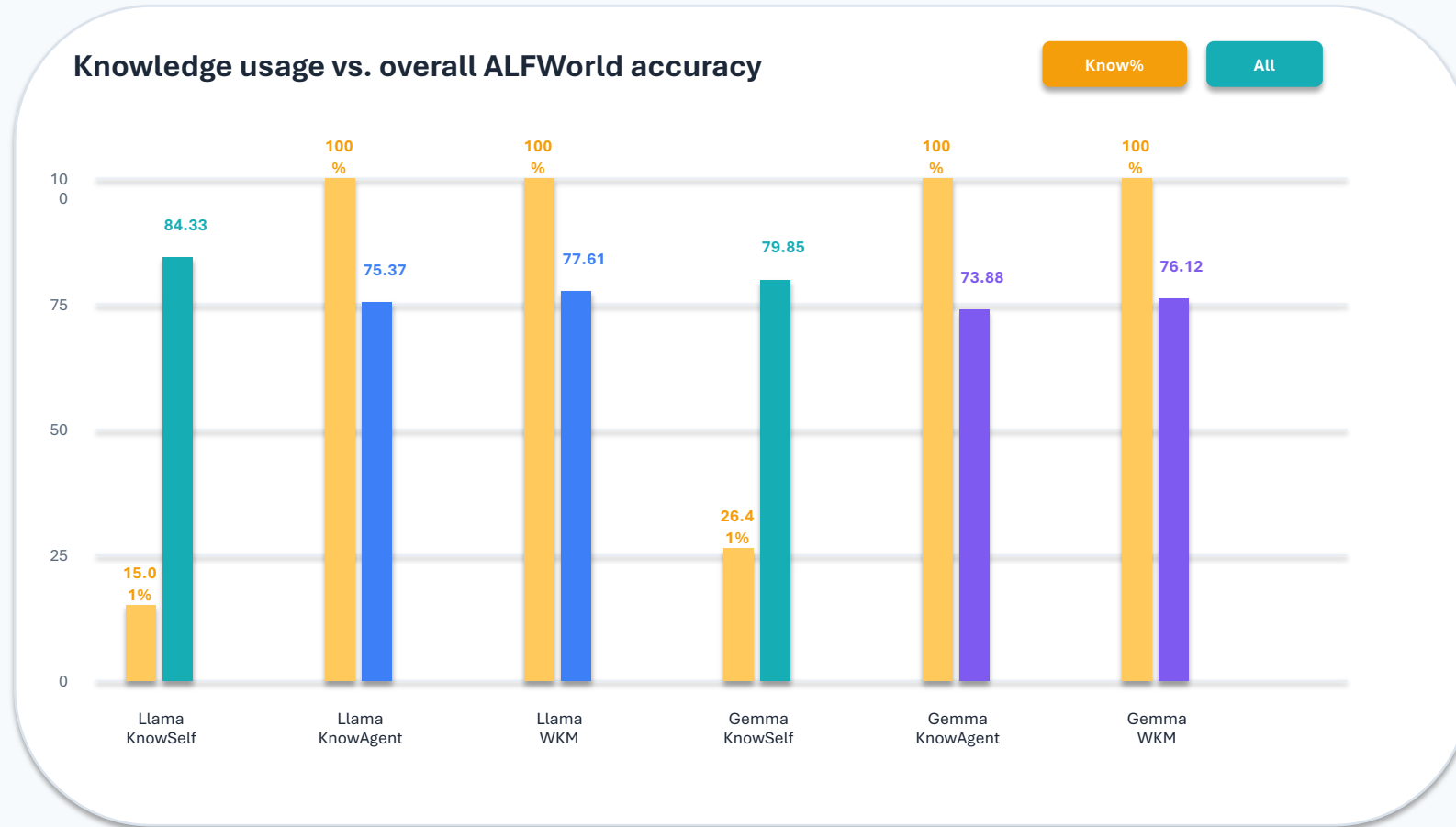
ExpeL · KnowAgent · WKM

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf	15.01%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf	26.41%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

Success rates (%) across six ALFWorld task types. Know% = proportion of steps using external knowledge.

Minimal Knowledge, Maximum Impact

KnowSelf uses selective knowledge queries (~15–26%) while matching or beating full-knowledge baselines.



Efficiency interpretation

- Llama-8B KnowSelf: 84.33% with 15.01% knowledge.
- Gemma-2B KnowSelf: 79.85% with 26.41% knowledge.
- Full-knowledge KnowAgent/WKM use 100% knowledge yet score lower on both backbones.
- GPT-4o ExpeL also uses 100% knowledge and reaches 79.85% overall.

Selective retrieval changes knowledge from a default dependency into a controlled cognitive resource.

Key Takeaways: Self-Awareness Is the Missing Piece

1

Indiscriminate knowledge injection is wasteful and can be suboptimal.

2

Agents can learn situational self-awareness via special tokens and heuristic labels.

3

Llama-8B KnowSelf reaches 84.33%, above GPT-4o ReAct at 76.12% and GPT-4o ExpeL at 79.85%.

4

Only 15–26% knowledge usage is needed versus 100% for competing knowledge-heavy methods.

5

The paradigm scales down: Gemma-2B KnowSelf reaches 79.85%, matching GPT-4o ExpeL.

Future directions

Richer self-awareness signals, lower-cost retrieval policies, broader environments beyond ALFWorld, and tighter integration with planning-time uncertainty.